S.A. ENGINEERING COLLEGE- CHENNAI-77
(Autonomous- Institute Level Research Centre, Affiliated to Anna University)
DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

AD1703A PROJECT PHASE - I

BATCH NUMBER : AD33

AI Based Invoice Analyzer using CRM Automation

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Abstract

In today's digital business environment, organizations handle a vast number of invoices daily, making manual invoice processing inefficient, time consuming, and error-prone. The proposed AI-Based Invoice Analyzer Using CRM Automation system addresses these challenges by integrating Artificial Intelligence (AI), Optical Character Recognition (OCR), and Natural Language Processing (NLP) to automate the extraction, validation, and management of invoice data. It extracts

Scope & Motivation

- Address challenges in manual invoice processing and disconnected CRM systems
- Reduce human errors and operational delays in invoice handling.
- Develop an AI-powered, end-to-end system for invoice analysis, CRM matching, upsell recommendations.

- Missed upselling opportunities due to lack of intelligent suggestions.
- Existing tools only solve isolated parts of the workflow.
- A unified AI solution can automate, accelerate, and optimize invoice-driven operations for better business outcomes.

Introduction

The AI-Based Invoice Analyzer Using CRM Automation aims to revolutionize invoice processing by leveraging advanced Artificial Intelligence (AI) and Machine Learning (ML) techniques. The system uses Optical Character Recognition (OCR) to extract text and key data fields such as invoice number, date, total amount, and vendor details from various formats like PDFs or scanned images. The extracted data is then validated and automatically matched with existing Customer Relationship Management (CRM) records to ensure consistency and prevent duplicate or fraudulent entries.

Literature Survey

S. no	Title of the Paper	Year	Journal/ Conference	Inference
1.	Integrating AI into CRM Systems for Enhanced Customer Retention	2025	Journal: Journal of Information Systems Engineering and Management 10(45s):474-489, DOI:10.52783/jisem.v10i45s.8892	AI (ML & NLP) boosts CRM automation and personalization, with generative AI poised for autonomous management despite data and adoption challenges.
2	An Efficient Deep Learning-Based Approach to Automating Invoice Document Validation	2025	Conference: National Institute of Applied Sciences and Technology(INSAT) University of Carthage Tunis, Tunisia, DOI:10.48550/arXiv.2503.12267	A DL pipeline with a new dataset uses LayoutLMv3 for accurate stamp/signature detection, overcoming distortions and imbalances for faster, error-free invoice processing.

Literature Survey

S. no	Title of the Paper	Year	Journal/ Conference	Inference
3.	Enhancing Customer Relationship Management in ERP Systems through AI: Personalized Interactions, Predictive Modeling, and Service Automation	2024	Journal: International Journal of Science and Research (IJSR) 13(3):892-897,DOI:10.21275/SR24313021820)	AI-integrated ERP-CRM boosts personalization, efficiency, and real-time decision-making, addressing Indian market gaps while tackling privacy and ethical concerns.
4.	FinTech - Automatic Payment Process in the ERP System	2024	Journal: International Journal of Computer Trends and Technology (IJCTT72(1):99-103, DOI: 10.14445/22312803/IJCTT-V72I11P116	ERP with AI, cloud, and blockchain automates finance, boosts security, and scalability, with triple-entry accounting aiding fraud prevention.

Literature Survey

S. no	Title of the Paper	Year	Journal/ Conference	Inference
5.	ExTTNet: A Deep Learning Algorithm for Extracting Table Texts from Invoice Images	2024	Journal: arXiv:2402.02246, https://doi.org/10.48550/arXiv.2402.02246	DocILE benchmarks AI for extracting key data from structured documents, providing annotations and promoting layout-aware models using visual, spatial, and textual cues.
6.	An AI-Based Customer Relationship Management Framework for Business Applications	2024	Journal: International Journal of Intelligent Systems and Applications in Engineering (IJISAE) – Volume 12, Issue 12s, 686-695.	An AI-driven CRM framework leverages ML and analytics for personalization, cost reduction, and profitability, while tackling data challenges.

Literature Survey

S. no	Title of the Paper	Year	Journal/ Conference	Inference
7.	An Overview of Data Extraction from Invoices	2022	Journal: IEEE Access PP(99):1-1 DOI:10.1109/ACCESS.2024.3360528	OCR, NLP, and graph-based ML extract invoice data despite layout challenges, enabling adaptable, low-intervention document management.
8.	Automatic Generation of a Custom Corpora for Invoice Analysis and Recognition	2019	Journal: 2019 International Conference on Document Analysis and Recognition Workshops (ICDARW) DOI: 10.1109/ICDARW.2019.60121	An invoice generator produces annotated datasets to solve data and privacy issues, using GCN for entity extraction across diverse layouts to improve AI recognition.

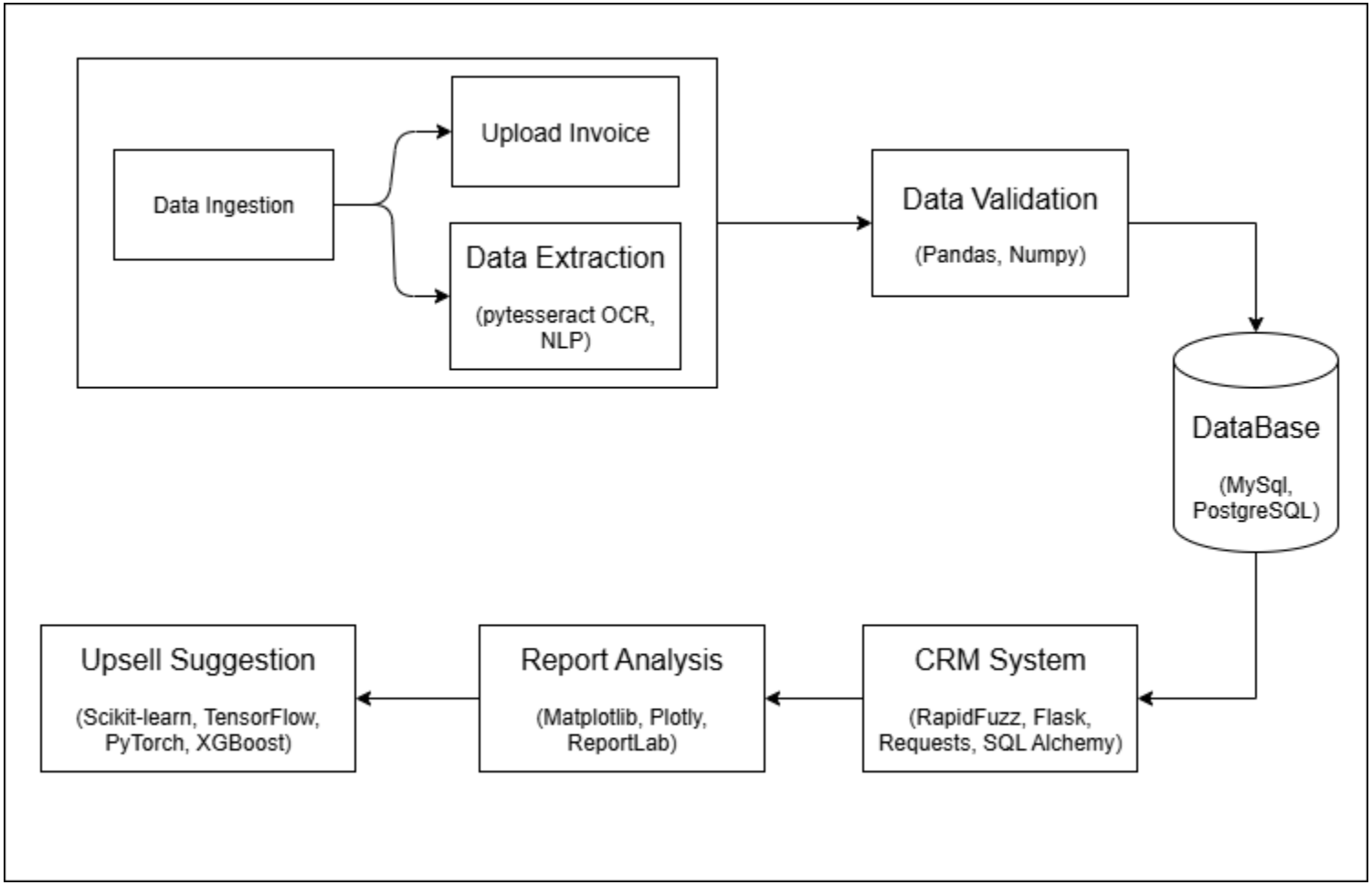
Objectives

- To develop automating invoice data extraction using Optical Character Recognition (OCR)
- Match vendor data with CRM records using fuzzy logic.
- Suggest upsell products using AI-based recommendation rules.
- Build a user-friendly web application to manage the full workflow.
- Reduce manual work and improve operational efficiency.
- Minimize manual data entry errors and improve operational efficiency in financial processing

Problem Statement

Manual invoice processing is time-consuming, prone to errors, and lacks integration with CRM systems. Existing tools handle only isolated tasks like OCR or CRM, but no unified system automates the full workflow including invoice analysis, customer matching, upsell suggestions. This leads to inefficiency, lost sales opportunities, and high operational costs.

Proposed Work-Block Diagram



Proposed-Novel Idea

Design an AI-based web application that seamlessly automates the entire invoice processing pipeline:

- Automatically scans and extracts data from invoices using OCR.
- Matches customer/vendor details with existing CRM records using intelligent matching.
- Provides personalized upsell suggestions based on previous transactions.
- All operations are managed through a user-friendly web application, eliminating manual effort.

Proposed-Modules

- Data Ingestion Module
- Data Validation
- CRM Matching Module
- Report Analysis
- Upsell Suggestion Module

Proposed-Module Description

1. **Data Ingestion**

- Collects invoices from multiple channels: email attachments, document uploads, scanned copies, and API integrations.
- Uses OCR (Optical Character Recognition) to extract invoice details such as invoice number, date, vendor name, line items, quantity, unit price, and total amount.
- Converts unstructured documents into a machine-readable format for downstream processing.

2. **Data Validation**

- Ensures extracted data is correct, complete, and consistent.
- Checks for:
  - Missing or incomplete fields.
  - Incorrect formats (e.g., invalid dates, wrong currency codes).
  - Duplicates and mismatched totals.
- Applies business validation rules so only accurate data moves forward, reducing processing errors.

Proposed-Module Description

3. **CRM Matching Module**

Uses fuzzy string matching (via RapidFuzz or similar) to compare extracted vendor names with existing CRM customer records, ensuring correct association even with minor mismatches or spelling variations.

4. **Upsell Suggestion Engine**

Analyzes customer's past purchase data to generate AI-based upsell recommendations. This module helps improve business sales by suggesting relevant or complementary products.

Software Requirements

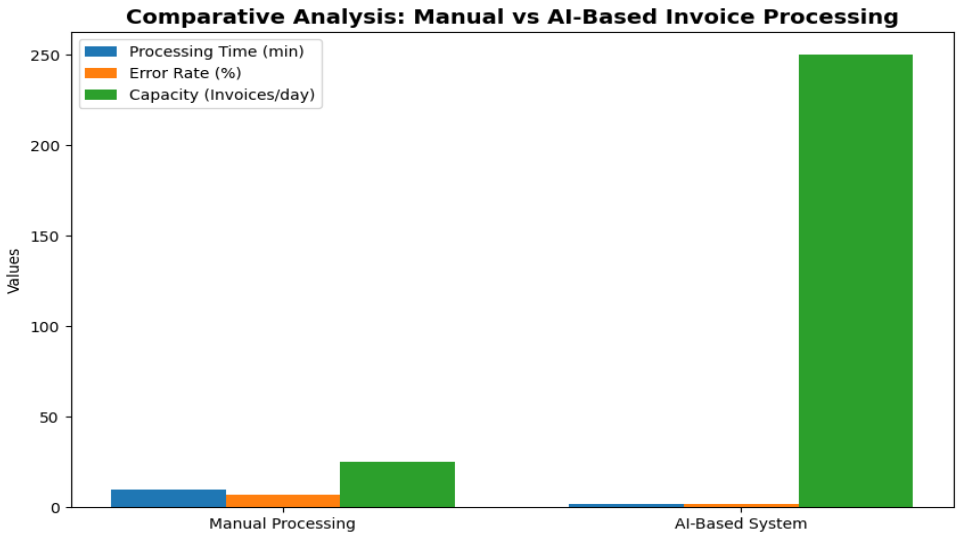
- **Operating System**: Windows 10 / Linux / macOS
- **Programming Language**: Python, JavaScript (ReactJS)
- **Libraries/Frameworks**:
  - Flask (Backend API)
  - Tesseract OCR (Text Extraction)
  - RapidFuzz (Fuzzy Matching)
  - Pandas, Regex, JSON (Data Processing)
  - ReactJS (Frontend UI)
- **Database**: PostgreSQL / MySQL
- **Development Tools**: VS Code, Postman, Git
- **Browser**: Chrome / Edge for frontend access

Hardware Requirements

- **Processor**: Intel i5 or equivalent (Minimum)
- **RAM**: 8 GB or higher
- **Storage**: Minimum 5 GB free (for dataset and model)
- **Graphics**: Not required (unless using GPU-based OCR/ML later)
- **Internet**: Required for data sync, updates, and API testing

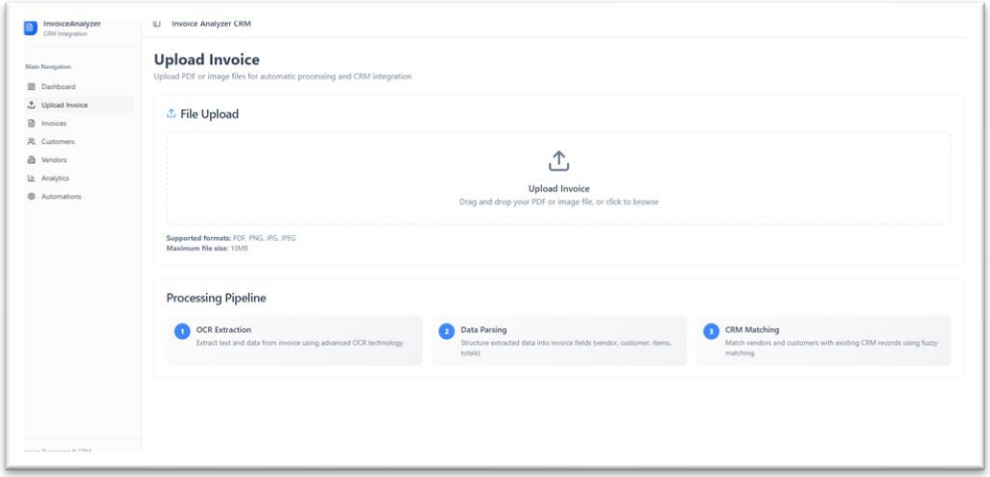
Result

- To demonstrate the effectiveness of the proposed system, a comparative analysis was carried out between manual invoice processing and the AI-based automated system. Manual invoice processing is time-intensive, inconsistent, and error-prone. The AI system drastically reduces processing time and improves accuracy. Automation ensures scalability and transparency, allowing organizations to handle bulk invoices efficiently. The system improves vendor relationships by ensuring timely approvals and payments

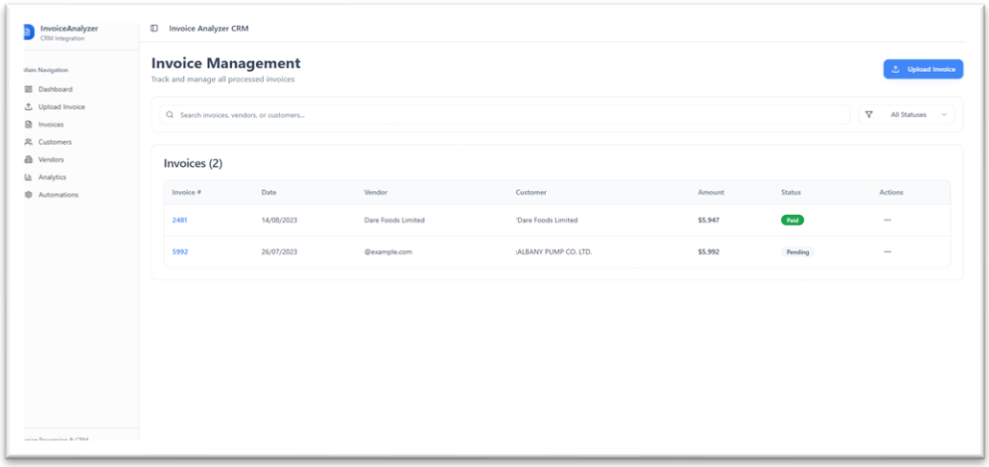


Snapshots

Upload Invoice:

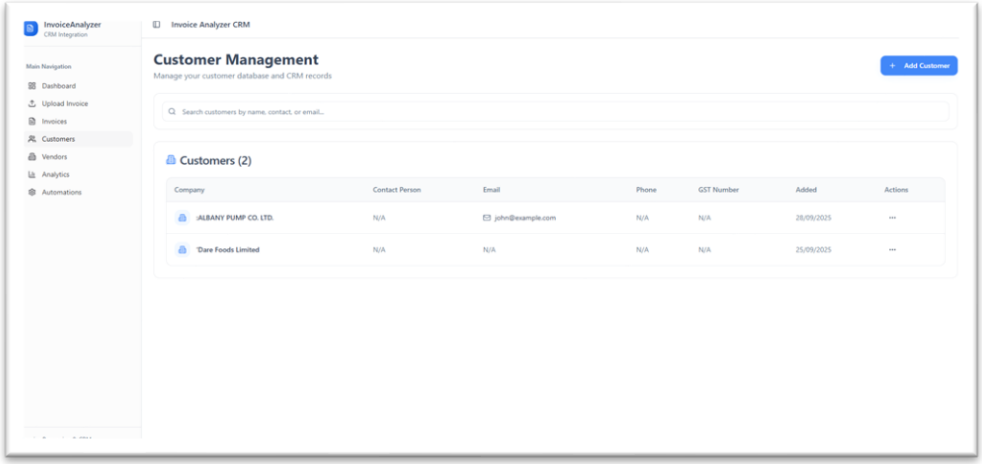


Invoice Dashboard:

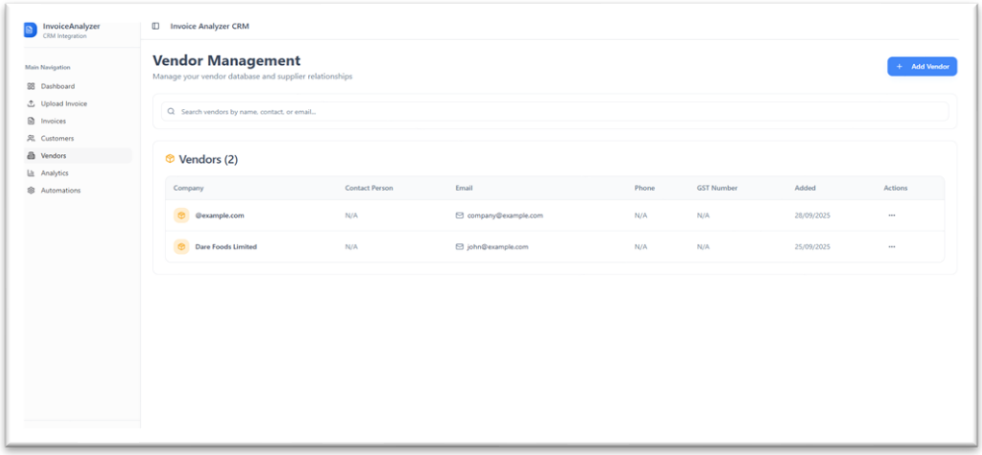


Snapshots

Customer Dashboard:

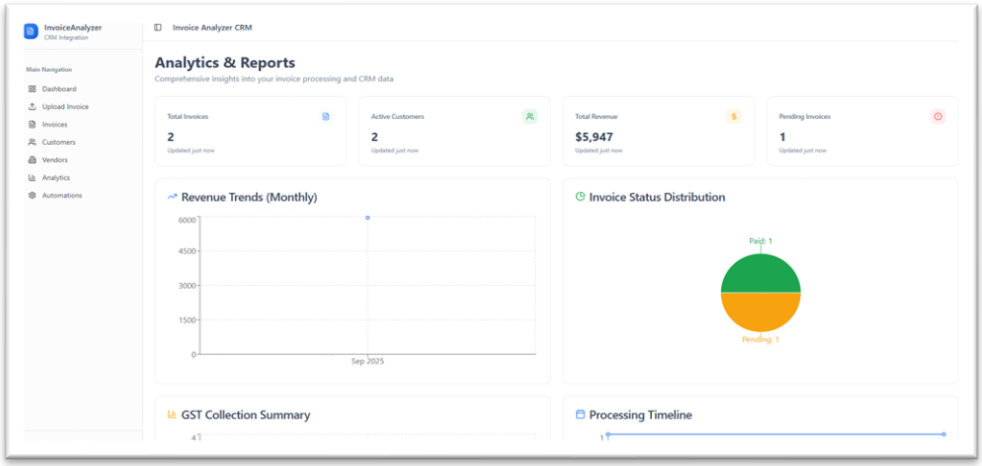


Vendor Dashboard:

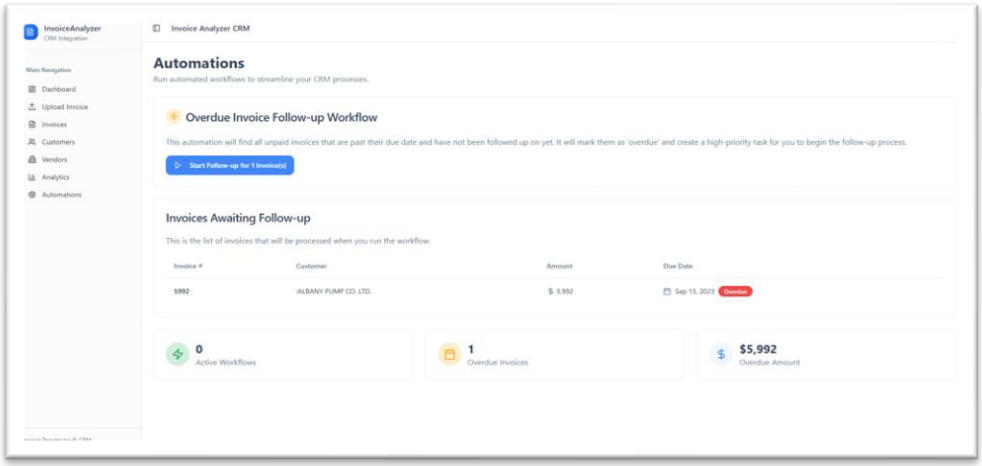


Snapshots

Analysis Dashboard:



Upsell Suggestion Dashboard:



Conclusion

The AI-Based Invoice Analyzer with CRM Automation effectively streamlines invoice processing through the use of OCR, NLP, and Machine Learning technologies. It reduces manual workload, minimizes human errors, and speeds up data entry and validation processes. The system accurately extracts key invoice details such as invoice number, vendor information, dates, line items, and total amounts. Performance evaluation shows high Accuracy, Precision, Recall, and F1-Score values, all exceeding 95%. Compared to manual methods, the AI-based system offers faster processing and lower error rates. Its seamless CRM integration ensures real-time updates of financial records and improved decision-making. The solution also helps reduce operational costs and enhances analytical capabilities

Future Enhancement

While the current system demonstrates strong performance, there are opportunities to further improve and expand its capabilities:

- **Integration with CRM System:**
  Beyond CRM, direct integration with ERP tools (SAP, Oracle NetSuite, Tally etc.,) for end-to-end financial automation.

- **Fraud Detection Module:**
  Incorporate AI models to detect duplicate invoices, anomalies, or fraudulent billing patterns.

- **Mobile App Support:**
  Allow users to upload invoices via a mobile app (camera-based scanning), making the system more accessible.

- **Adaptive Learning & Continuous Improvement:**
  Implement self-learning models that improve accuracy as more invoices are processed over time.

- **Advanced Analytics & Dashboard:**
  Provide detailed financial insights, trends, and predictive analytics through interactive dashboards.

References

[1] Kumar Subham, “Integrating AI into CRM Systems for Enhanced Customer Retention”, Journal of Information Systems Engineering & Management 10(45s): 474-489, DOI: 10.52783/jisem.v10i45s.8892, April 2025.

[2] Aziz Amari, Mariem Makni, Wissal Fnaich, Akram Lahmar, Fedi Koubaa, Oumayma Charrad, Mohamed Ali Zormati, Rabaa Youssef, “An Efficient Deep Learning-Based Approach to Automating Invoice Document Validation”, National Institute of Applied Sciences and Technology(INSAT) University of Carthage Tunis, Tunisia, DOI:10.48550/arXiv.2503.12267, March 2025.

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[5] Adem Akdogan, Murat Kurt, “ExTTNet: A Deep Learning Algorithm for Extracting Table Texts from Invoice Images”, arXiv:2402.02246, https://doi.org/10.48550/arXiv.2402.02246, February 2024.

[6] Rashi, Dr Biplab, Kumar Biswal, Srinivasa Rao Yadavalli, “An AI-Based Customer Relationship Management Framework for Business Applications”, International Journal of Intelligent Systems and Applications in Engineering 12(12): 686-695, January 2024.

[7] Thomas Saout, Frederic Lardeux, Frederic Saubion, “An Overview of Data Extraction from Invoices”, IEEE Access, PP(99):1-1 DOI: 10.1109/ACCESS.2024.3360528. January 2024.

[8] Jerome Blanchard, Yolande Belaid, Abdel Belaid, “Automatic Generation of a Custom Corpora for Invoice Analysis and Recognition”, 2019 International Conference on Document Analysis and Recognition Workshops(ICDARW), DOI: 10.1109/ICDARW.2019.60121, September 2019.